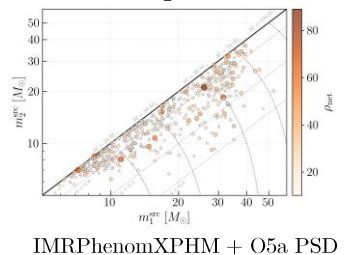


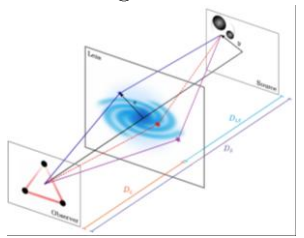
Stage 1 (Ch. 5): Population Generation

Event
Generation

Source &
Lens Population



Lensing Simulation



Detectability filter

152k candidates drawn

500 detectable ($\rho_{net} \geq 10$)

309 pass duty cycle (0.70)

157 final events
(≥ 2 img, $\rho_{net} \geq 10$)

Stage 2 (Ch. 6): Pipeline setup

Templates

- Single-image and joint .ini templates + SLURM script templates

Lenskyloc_inj_hanabi_single.ini

Lenskyloc_inj_hanabi_joint.ini

File generation

Generator_inis_lenskyloc.py populates priors and injection params per event

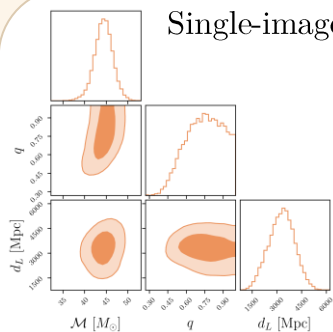
Output files

Two single-image .ini files and one joint .ini file per event + SLURM scripts for post-processing (result merging, SNR recovery, sky map generation)

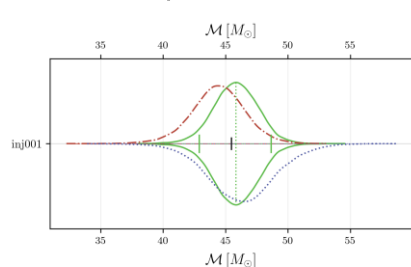
Stage 3 (Ch. 6): Parameter estimation

BILBY

Single-image PE

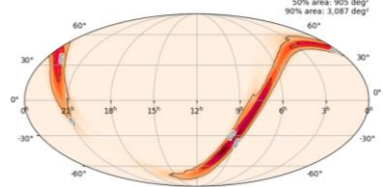


Joint PE

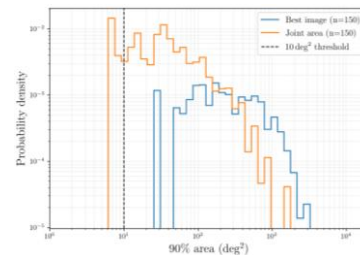


Stage 4 (Ch. 7): Sky localization

Posterior to skymap



90% credible area



Single vs joint

